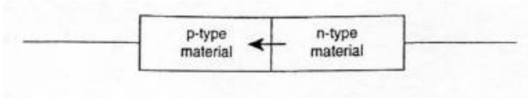
5 (a)



Electrons, the majority charge carrier in the n-type material, diffuse over to the p-type material.

(b) The majority charge carriers in the n-type material are negatively charged electrons while that of p-type material are positively charged holes.

When the p-type and n-type materials come into contact, electrons in the conduction band diffuse from the n-type to the p-type material. At the same time, holes in the valence band diffuse from the p-type to the n-type material.

The recombination of electrons with holes on both sides give rise to a region depleted of electrons and holes, resulting in a depletion region.

The n-type side of the depletion region is positively charged due to immobile positive ions while the p-type side is negatively charged due to negatively charged ions.

These region of immobile fixed charges set up an electric field directed from the positively charged n-region to the negatively charged p-region, thus preventing further diffusion of charges in either direction.

(c) To increase the width of the depletion region, the p-n junction has to be in reverse bias. This means the positive terminal of the battery applied to the n-type material, and the negative terminal of the battery applied to the p-type material.